From Raw Data to Features

INDICATORS ARE FILTERS

The market as audio

Traditional trading methodologies view the market as a spreadsheet

- This is **incorrect**
- Instead, we can picture the market in our minds as a noisy audio file

So what?

The Engineering Metaphor: Market as a Waveform

An audio file is simply *air pressure* measured over time

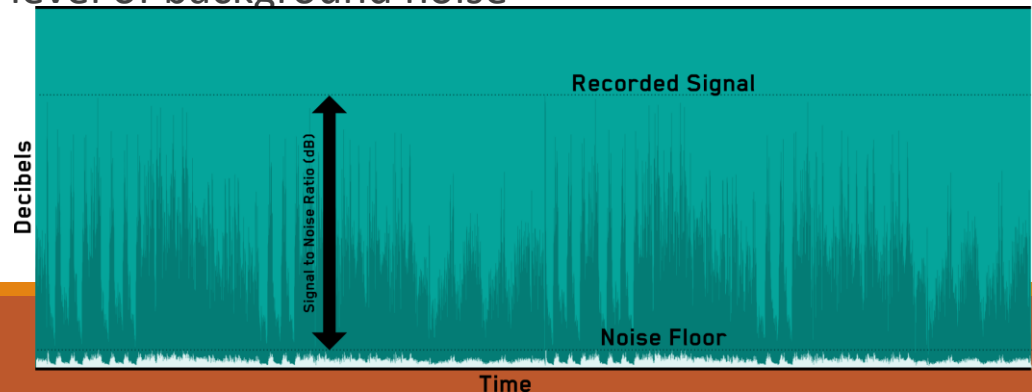
- We use a microphone to capture sound waves (air pressure changes) and we convert them into electrical signals which are then amplified and stored as data (0 or 1)

When we record an audio file *almost all* recordings contain **noise**

- Noise is an unwanted sound that interferes with our primary signal
  - Ranges from constant electronic hums and background ambiance to accidental sounds that degrade quality
  - Appears as random, unstructured energy above the primary signal
  - It's essentially anything you don't want to hear that *masks* the details of our primary signal

When we consider the quality of audio, an important term is our **signal-to-noise ratio**

- This is the measure of unwanted audio signal compared to our level of background noise



The Stock Market

The stock market is simply **price** measured over **time**

In finance, we capture thousands of **ticks** per second to recreate the market

Just like audio, raw market data is *dirty*

Sources of noise in the stock market data

- High-Frequency Trading (HFT) bots fighting over fractions of a penny.
- Liquidity gaps, "Fat Finger" trades
- Emotional panic
  - FOMO
  - Market Headlines

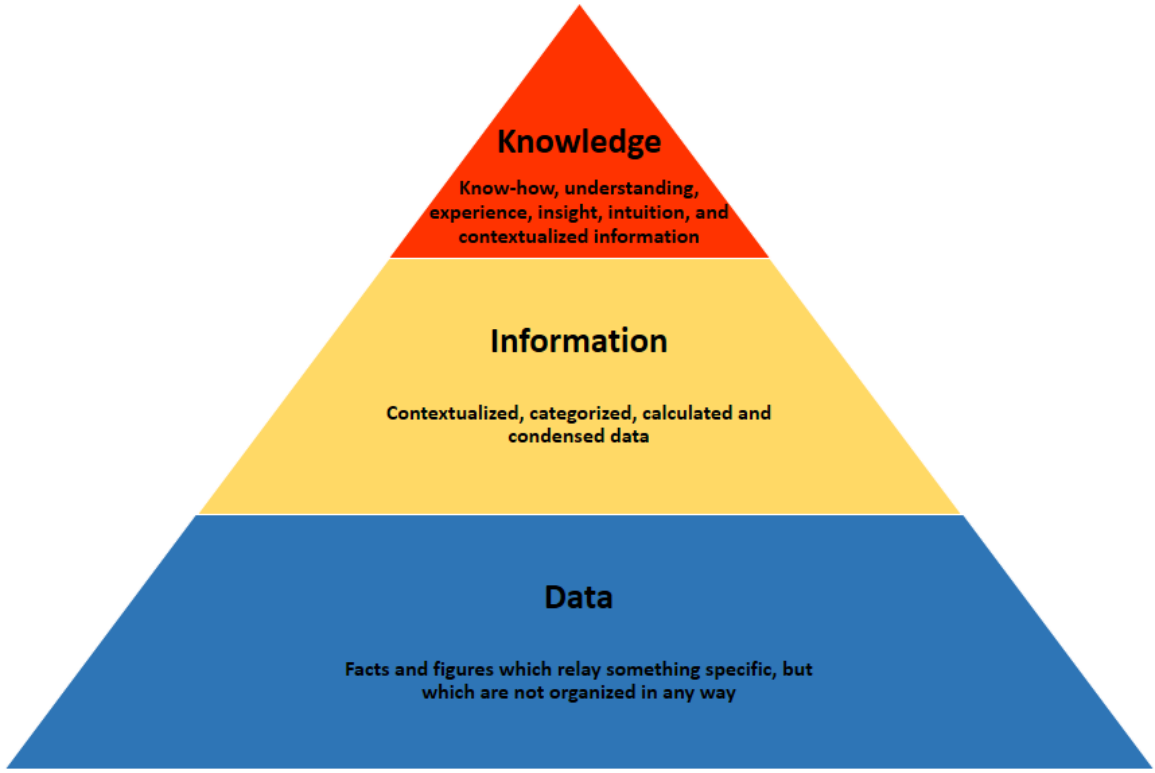
The Signal Processing Problem

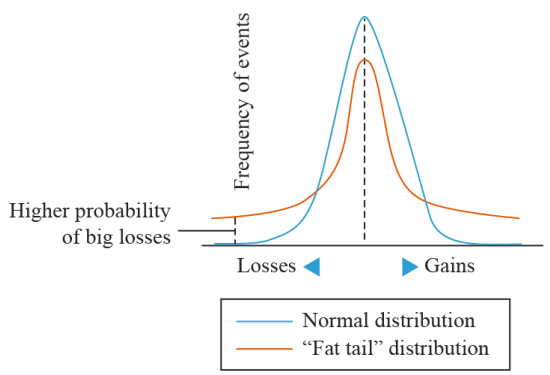
Ticks are high-frequency, chaotic, individual moments of action in a noisy market
- They are the *raw audio* of the market

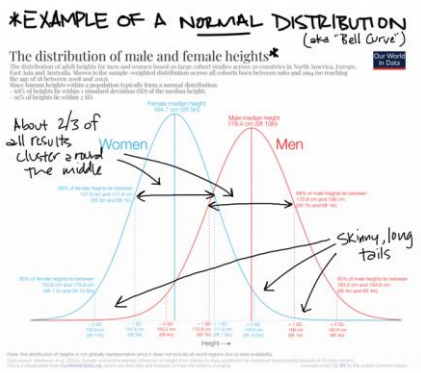
They convey **data**
- Data is **not** information
  - This is a **massive** distinction.

The distinction
- **Data** is a raw stream
  - $150.01, $150.05, $149.95
- **Information** is an *actionable state*
  - The price is rising
- **Noise**
  - Random variance that has no predictive value

We want to go past analyzing charts
- We want to design filters to reject Noise and pass signals







The Nature of Noise

The Engineering Assumption (The Bell Curve)

- In standard engineering, noise is *gaussian*
  - Normally distributed
- Events cluster around the average
  - Extreme outliers are mathematically impossible

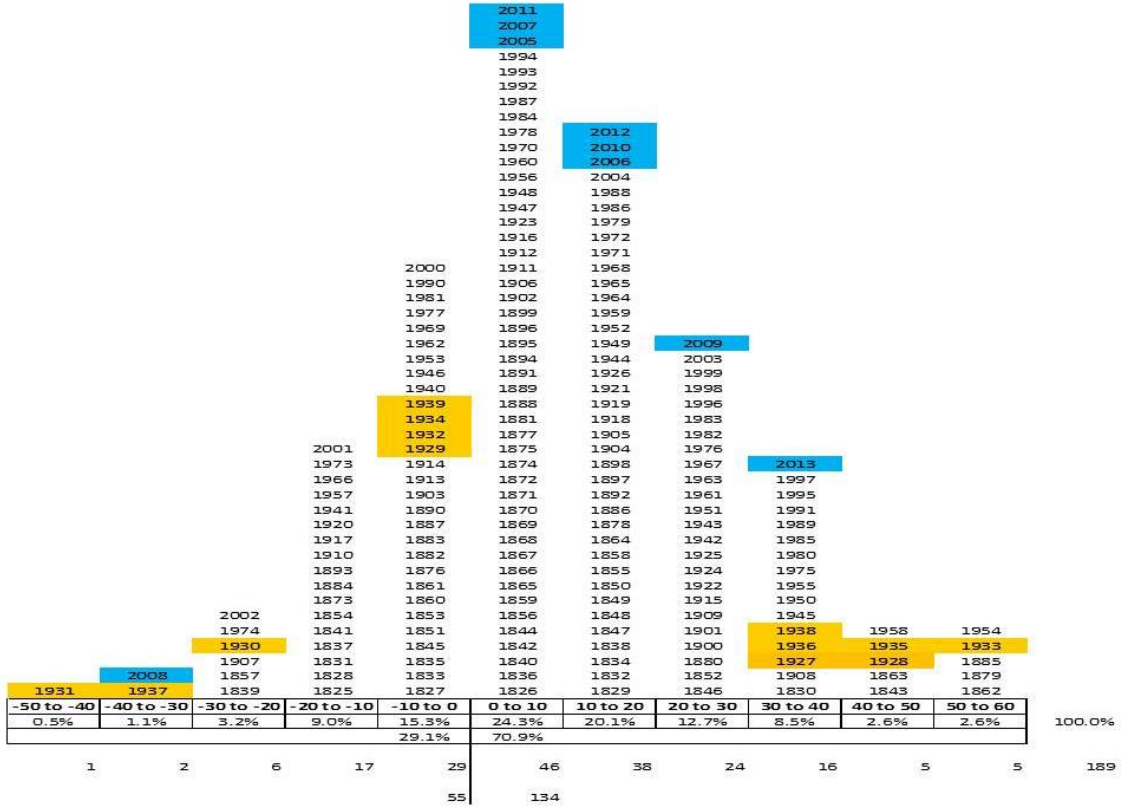
The Market Reality

- Markets follow a **Power Law** (Pareto Distribution)
  - 80% of outcomes are due to 20% of causes
- Extreme events are not outliers; features of the system

The *black swan* failure method

- If we try to calculate risk without trying to understand noise, we build risk
- A *stop-loss* could trigger, a single bad day of trading erases months of profit.

Histogram showcasing average historical returns of the S&P 500 since 1825



Signal-to-Noise Ratio (SNR)

The Market is 90% Noise, 10% Signal

In radio:

- SNR = $\frac{SignalPower}{NoisePower}$

In finance:

- SNR = $\frac{Trend(DirectionalDrift)}{Volatility(Variance)}$

The mathematical Reality:

- Price = Trend + Noise
- Most novice traders lose money because they treat the Noise as if it were the Trend

Our goal

- Use indicators to not predict the future, but to separate the sine wave from the static
- We need mathematical tools that can suppress the jagged spikes (Noise) so we can see the sine wave (Trend) clearly.

The Effect of Noise

Let’s dive into this a bit

The market is 90% noise, 10% *true shift* in value

Think about what this does to an algorithm

If we build a system that tries to react to every price change

- We are effectively gambling

You let the 90% of noise dictate our wallet and results

- **most** novice traders lose money not because they are *unlucky* but because they are mathematically treating the noise as if it was the trend

Our job this semester is to **not** predict the future

We hope to build systems and filters to block noise, understand true market trends, and thus make better informed decisions

Simple Moving Average

A Simple Moving Average will be our first attempt at managing noise

We can do this by trying to **smooth** out our graph

- It is designed to **block** high-frequency noise (ticks/spikes) and **pass** low-frequency trends
- By taking the average of the last *N* periods, individual outliers(positive and negative) cancel each other out

The Transformers

- **Input:** Volatile, ticks
  - Chaotic price action
- **Output:**
  - A smooth, tradeable line
    - In theory….but what does the SMA introduce?

SMA: The low pass filter

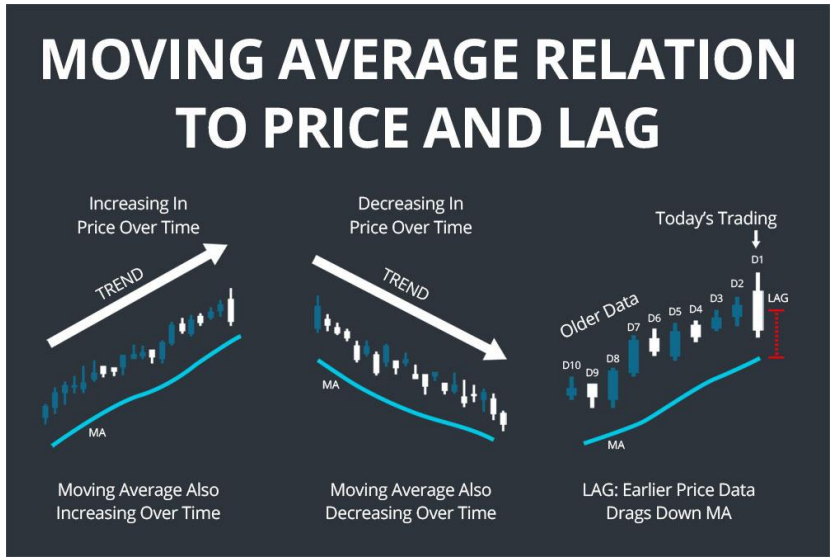
Let's once again, revisit our simple moving average

- The goal is to remove high-frequency noise(ticks) to reveal the low-frequency trend

We treated a moving average as a single calculation

- Price > average
  - Buy signal
- We also ultimately introduced *lag*
  - You cannot smooth the present without using the past.
  - The green trend line happens after the orange peaks
    - This is the engineering price we pay for clarity

So how do we calculate it?



Let’s find it!

On canvas, I have uploaded a notebook that does this for us, let’s take a look

SMA discussion

discuss with your neighbor

What does it do well?

What does it not do well?

Fighting Lag — The Exponential Moving Average (EMA)

The SMA Flaw: “Democracy”

- In a simple moving average, the price from 19 days ago has the **exact same vote** as the price today.
- This is an *engineering failure* as data from 3 weeks ago is *stale*

The EMA Solution

- We introduce EMA to encode *recency bias*
- We add a multiplier to give more weight to more recent data points
- We *forget* the past exponentially faster than the SMA

The Result

- The EMA signal is shifted closer to *now*

The trade off

- More sensitive to noise but catches the trend change quicker

How EMA is calculated

$$EMA = K \times (\text{Current Price} - \text{Previous EMA}) + \text{Previous EMA}$$

K: The weighting factor for the EMA.

K = 2/(n+1)

Where:
n = the selected time period

For first time calculations, the Previous EMA is the initial EMA, which is simply an average of all prices over 'n' number of periods (a Simple Moving Average):

'n' period sum of prices / 'n' number of periods

In other cases, the Previous EMA refers to the EMA from the previous trading day.

Our goal is to *add memory to a machine*

Unlike SMA that needs 200 or more past prices, the EMA only needs two numbers

- The Current Price (what is happening *now)*
- The previous EMA (What happened *before*)
  - Assuming it has been calculated
  - It's recursive in nature
    - Because today's value is built on yesterday's value (which was built on the day before...)
    - The EMA theoretically "remembers" data from the very beginning, just with very tiny weight.

We take a *chunk* of todays price modified by a weight

- The weight is our modifier that determines how much we care about the new price vs old history

EMA Caveats

Just like SMA's an EMA can be calculated over multiple different timelines

- EMA's taken over shorter periods of time are more sensitive to prices
  - A 21 day EMA follows *live* prices more closely compared to a 100 day EMA

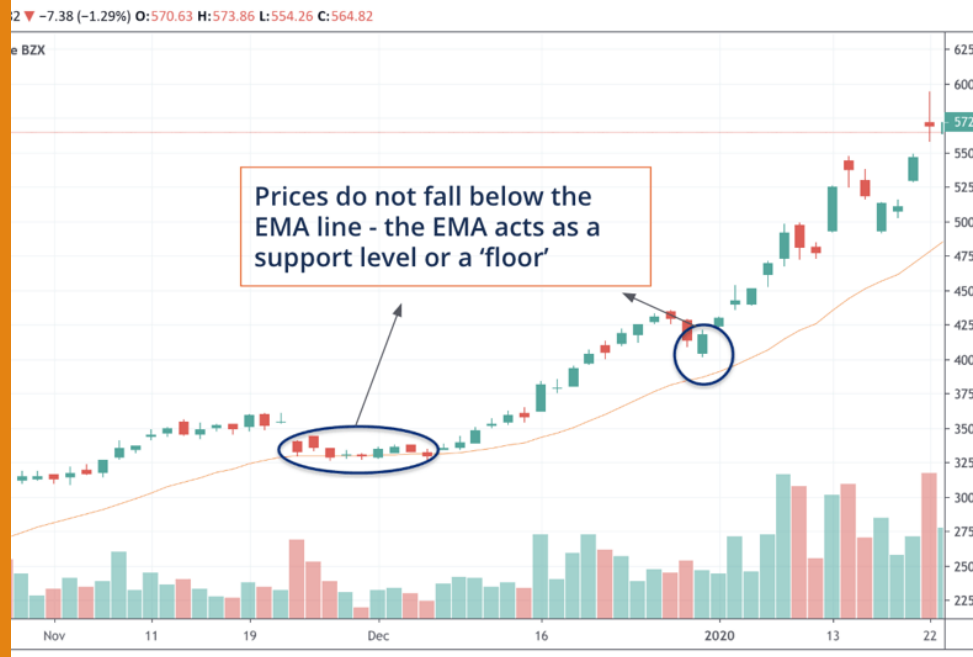
From Data To Information

EMA's highlight trends

- Highlighting and identifying price trends is the principal goal of an EMA
- By identifying price directions, the EMA allows investors and traders to spot buying and selling signals
  - Price is above the EMA line, likely to rise
  - Price under the EMA line, likely to fall

EMA builds *support* and *resistance* bands

- Support levels are known as *floors* they act as a limit for how far prices are expected to fall
- Conversely, Resistance levels can act as *ceilings* in downtrends



EMA vs SMA

We understand that when making the EMA and SMA charts, we are **abstracting** data away

- This helps reduce noise and allows us to see trends

There is no getting around the fact that we **lose** data when doing this

None of the moving average calculations is a better indicator than the other

- They both serve their own purpose
- They are both useful

Although an EMA is a more accurate representation of recent price movements and helps identify trends quicker

- It **also** experiences more short-term fluctuations compared to a SMA

Our First Trade(signal)

THERE ARE MANY – LET’S LEARN JUST ONE FOR NOW

Measuring Internal Momentum(RSI)

Position vs Energy

- **Price** tells us *where* the stock price is (e.g. $150)
- **Relative Strength Index (RSI)**: tells us *how much energy* it used to get there

The law of Exhaustion

- If a runner sprints 100 meters in 9 seconds – they are pretty fast(high price)
  - As a side effect, they are **exhausted** (RSI is high)
  - They can't sprint another 100 meters immediately

Why this helps filter noise

- A sudden $10 price spike looks like a *breakout*
- An RSI reveals that the move required 100% of the buyer's energy
  - The move is likely *noise*
    - *Panic, FOMO* and thus not sustainable
- The inverse, $10 price run could also be true.

The Problem of Scale

The Engineering Challenge

Q: If a stock moves up $5.00, is that a big move?

- A: It depends on the denominator
  - For a $5.00 stock, a $5.00 move is a 100% increase(massive signal)
  - For Berkshire Hathaway($700,000) a share, a $5 move is a rounding error
  - If we build a trading bot using hard coded values, (e.g. **IF price_move > $5 THEN BUY)** then your bot fails the moment you swap classes

Q: How do we build a single algorithm that works for any asset class without hard-coding dollar values

- A: We cannot use absolute dollars, we must use **relative ratios**
  - A relative ratio is a measurement that compares a value to its own history rather than an arbitrary external standard
  - **How RSI uses it:** Instead of looking at "Dollars Gained," RSI calculates the ratio of Average Gains to Average Losses over a set period (usually 14 periods)

Relative Ratios and Normalization

This involves **normalization**

- **Normalization:** is the mathematical process of compressing values from an infinite range (e.g., price can technically go to infinity) into a strictly bounded range (e.g., 0 to 100)

Using Normalization, we can build *signals* that monitor and can *assess* assets **regardless** of the price tag

- We can label 0 as our **absolute floor**
  - Zero buying pressure, nobody wants it
    - Your grandfathers VCR collection
- We can label 100 as our **absolute ceiling**
  - Maximum buying pressure
  - Zero sellers left
    - Hotcakes

Q: Did anyone know a hotcake was a pancake?

The Zones (Reading the Thermometer)

Our goal with RSI is to measure the velocity of **directional change** change

$$\mathrm{RSI} = 100 - \frac{100}{1 + \text{Avg Gain}/\text{Avg Loss}}$$

Think about a car engine's tachometer(RPMs). You can drive in the red zone, but not for long before something breaks

We can break our normalized scale into *zones* that each convey different signals

The upper limit: overbought > 70

- The Signal: The asset is overheating. The velocity of the move is statistically unsustainable.
- The Physics: The buyers have expended almost all their available capital (energy) to push the price this high.
- Probability: The likelihood of Mean Reversion (price snapping back to the average) increases significantly

The Lower Limit: Oversold (< 30)

- **The Signal:** The selling pressure is exhausted.
- **The Physics:** Everyone who wanted to panic sell has already exited
  - There are no sellers left to push the price down lower

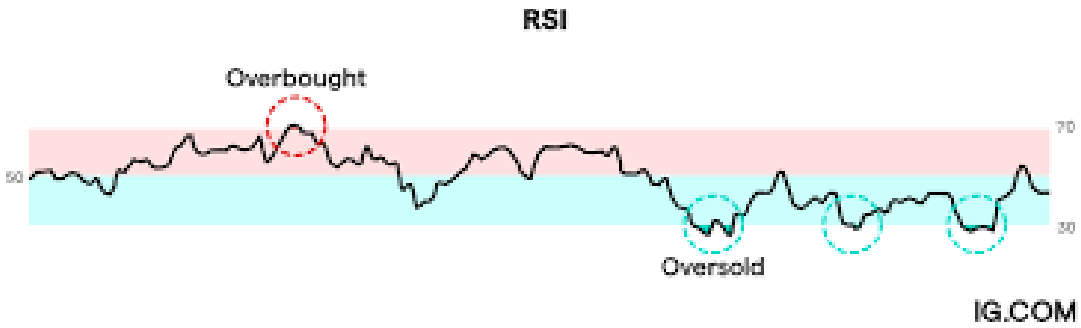
Q: This sounds perfect! > 70 SELL < 30 BUY! What do we think!?!

The Engineering Trap

Most beginners really think this

- > 70 ,sell!
- < 30, buy!

Indicators are powerful, but they lack wider **context**



Overbought does not mean sell

- It just means *high energy*

Say a powerful prospectus comes out for some company and FORD goes on a massive breakout

- A stock can stay “overbought” at 80 or 90 for *weeks* while the price doubles
  - If you sell to early, you lose out
  - If you short it, you lose money

RSI is a condition, not a trigger

- It helps us paint our environment, **not** the timing

From Analog to Digital (Logic Gates)

We can look at a chart and say: *that looks pretty cheap"*

Our bots **don't** have eyes or feelings(for now……)

- It cannot execute on "pretty cheap"
- It cannot execute on a **Float**(RSI=29.5)
- It can only execute on a **Condition**(Buy=True)

Digital systems have no nuance.

- They only have conditions.
- This makes them fast, but it also makes them rigid.

Parameter Tuning(The Bias Problem)

We must define a specific number where *analog* becomes *digital*

The Hardest Decision (Parameter Tuning):

- *Example:* **IF** RSI < 30.
- Why 30? Why not 29? Why not 31?
- Every threshold you pick creates a **Bias**. Too tight = Missed trades. Too loose = False signals.

A **Threshold** is the *exact* point where "safe" becomes "danger"

- A vampire can't enter a house unless their invited past the threshold(door)
  - …Or in our case, *wait* becomes *buy*

So, how do we go about picking this threshold?

- Reality: There is no "correct" number. The number 30 is not a *law of physics*

Parameter Tuning Continued

Scenario A: The Strict Gate (RSI < 20)

- The bot almost never trades
- Waits for the market to be absolutely destroyed before buying
  - Pros
    - High confidence, if the RSI is 19 a bounce is *almost* guaranteed
  - Cons
    - You miss about 90% of the good trades

Scenario B: The Loose Gate (RSI < 40)

- The bot trades constantly
  - Pros
    - You catch every dip
  - Cons
    - You buy “dips” that often keep dipping

Terms

Data Science Theory:

- Strict Threshold: High Precision (low risk), Low Recall (few trades)
- Loose Threshold: Low Precision (high risk), High Recall (Many Trades)

This idea of **parameter bias** is how greedy or picky algorithms are born

- Picky: nothing happens and you die of old age
- Greedy: The algorithm is too greedy, you go broke!

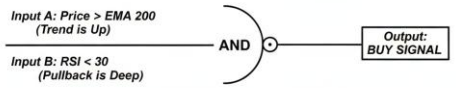
The "AND" Gate (Confluence)

The idea of **confluence** is that we never rely on a single sensor

- Sensors fail

This idea of redundancy in sensors and systems is not new

We use *logic gates* to combine sensors with different biases



The strategy (Trend following pullback):

- **Sensor 1 (EMA):** "We are in a Bull Market" (Permission to Buy).
- **Sensor 2 (RSI):** "The price is temporarily cheap" (Trigger to Buy).

The Logic:

- Signal = (Trend == BULLISH) AND (Momentum == OVERSOLD)

Why This Works:

- You are buying the dip (RSI) but only in the direction of the main current (EMA).

Our First (Novice) Strategy

The Logic

- Signal = (Price > EMA_200) AND (RSI < 30)

Condition A:

- Is the current price above the 200-day exponential moving average
  - If yes, bull market

Condition B:

- Is the relative Strength Index below 30?
  - If yes, oversold

If both are true, fire a buy signal!

In algorithmic terms, we have just create a novice way to encode the philosophy of **buying the dip**

Avoiding FOMO:

- If we only used EMA filter, we buy at the top of the bubble
  - The RSI filter forces us to wait for a pullback